

Emergency Response App — Cardiovascular Hospital

Product Overview

A mobile emergency response system used by hospital staff to receive cardiac alerts, understand patient context, and coordinate response during high-stress scenarios. The hospital already had a working emergency app built by another vendor.

The objective was not to rebuild it, but to simplify it for real emergency behaviour — especially during night shifts.

My Role

UX/UI Designer

Worked closely with:

- Product Manager
- Engineering team
- Nurse heads and ward coordinators

Focused on:

- Restructuring alert and communication flows
- Redesigning critical emergency surfaces
- Improving information hierarchy
- Aligning UI with Ionic / Material constraints
- Iterating through prototype walkthroughs with hospital staff

The Real Problem

In cardiac emergencies, the first 30 seconds determine response quality.

In the existing app:

- Alerts displayed vitals, history, chat, and actions together
- Nurses had to interpret data while deciding what action to take
- Communication and alerts were tightly mixed
- Dense layouts slowed recognition, especially at night

Nothing was technically broken.

But cognitive load was highest at the exact moment clarity mattered most.

Core Design Principle

Separate scanning from acting.

Users should first understand what’s happening.

Then decide.

Then act.

Not all at once.

Structural Shift

Instead of presenting everything simultaneously, the experience was divided into two clear layers.

1. Scan Layer (Context First)

Opening an alert shows a single, large alert card containing:

- Patient vitals (HR, BP, O₂) in enlarged typography
- Ward / room in high contrast
- Alert type and severity
- Time elapsed

Intentionally removed from this layer:

- Call buttons
- Chat interface
- Patient history

This screen exists purely for fast comprehension.

2. Action Layer (After Context)

Once the situation is understood, users can access:

- Call → Direct contact with on-call physician
- Chat → Team coordination
- Patient details → Full medical history

Actions are one tap away, but never competing with initial scanning.

This reduced hesitation and accidental interaction during high-stress moments.

Information Architecture Redesign

Previously, alerts and communication were blended into a single experience.

The system was reorganised into:

Alarm List

- Dedicated to emergency alerts
- Designed for speed and clarity

Communication Hub

- Group chats
- Message history
- Staff coordination

Users can move between chat and patient context without losing state, creating an interconnected experience without cluttering emergency surfaces.

Readability Under Pressure

Stakeholders repeatedly emphasised:

“It must be readable at 2 AM.”

Initial iterations increased font size slightly.

This was insufficient.

The adjustment:

- Increased overall card scale, not just typography
- Reduced information density
- Accepted vertical expansion since nurses handle one case at a time

Improvement came from layout scale and hierarchy — not decoration.

Constraints

- Workflow ambiguity for 1–2 months due to evolving AI discussions
- Mid-project stack shift from React + Material to Ionic
- Component limitations requiring reuse of system elements

Instead of redesigning the full visual system, effort focused on surfaces that directly affected emergency response behaviour.

Validation

- 3 prototype iterations
- 5 nurses
- 2 ward coordinators

Early prototypes were rejected for insufficient visual scale.

Later iterations improved alignment once discussions centred on speed, recognition, and error reduction.

Outcome

- Product shipped and went live
- Alert flow consistently met the sub-30-second response target
- Staff described the experience as more comfortable than Epic-like systems during emergencies
- Reduced hesitation before first action

The impact was behavioural — not feature-based.

Reflection

This project was not about adding functionality.

It was about reducing mental effort in the most critical window of use.

By restructuring the system around scanning before acting, the app became faster, clearer, and more dependable under pressure.